

KEEPSAKE

New Art Direction & Room Production Bible

Illustrated Nostalgia • Woodland 2.0 • Four-Room System • Procreate / Figma /
Blender / Mirage Tesseract / Three.js



Art-direction reference supplied by the project owner: Mirage Tesseract demo screenshot. Used here as a mood/composition reference.

Prepared September 2026 • Comprehensive planning document for the Keepsake project and Astra workflow

1. Executive Summary

Keepsake is evolving from a collection of functional 3D room prototypes into a deliberately art-directed memory environment. The goal is no longer simply to make the existing rooms more detailed. The goal is to establish a repeatable production system that makes every room feel authored, nostalgic, coherent, performant, and emotionally connected to the scrapbook.

Core creative thesis

The scrapbook tells the story of the user's life; the room tells the story of the user's relationship with those memories. Rooms are authored spaces, not room builders. Users personalize them through memories, artifacts, small expressive objects, seasonal traditions, and accumulated history rather than by replacing major furniture.

The chosen visual target is a middle ground called Illustrated Nostalgia: more dimensional and materially believable than a flat illustration, less literal than photorealism, and more carefully authored than a generic low-poly or asset-store room. The target combines softly stylized geometry, painterly or cohesive materials, cinematic lighting, restrained motion, tactile hero props, and a strong warm-versus-cool lighting language.

Decisions now considered locked

- Use authored camera viewpoints and smooth transitions as the primary navigation model. WASD/free-roam can be revisited later, but it is not a launch requirement.
- Keep a shared functional room skeleton across themes, while allowing each theme to reinterpret furniture, materials, lighting, and architecture.
- Eliminate major furniture swapping. Desks, chairs, beanbags, cabinets, and other composition-defining furniture are authored as part of each room.
- Preserve personalization through small decor, CRT treatment/content, stickers, books, photographs, window-sill objects, display-case artifacts, seasonal objects, and memory-driven details.
- Use Procreate-painted exterior scenes split into depth layers and presented as 2.5D parallax environments.
- Treat Blender as the authority for physical composition and placement; treat Three.js as the authority for behavior, interaction, dynamic state, and presentation.
- Use Mirage Tesseract as an optional production/prototyping tool for 2.5D motion, atmospheric loops, CRT content, and visual experimentation - not as the runtime room engine.
- Consolidate Woodland and Snowy Mountain into one seasonal Woodland room. Winter becomes a transformation of the familiar room rather than a separate destination.
- Reduce the room lineup to four distinct fantasies: Woodland Retreat, Beachfront, Neon City, and Sky Castle.
- Only load/render what is relevant to the current view, room, season, and interaction state.

What changes from the current prototype

Current tendency	New direction
Rooms accumulate runtime replacements and procedural systems	Rooms are composed as authored scenes with a strict art pipeline
Exterior worlds use substantial runtime geometry/shaders	Exterior worlds become painterly 2.5D illusions whenever

	full 3D is unnecessary
Furniture can be swapped broadly	Major furniture is fixed per theme; personalization moves to smaller memory-bearing objects
Generic front/left/right camera inheritance	Each room receives its own approved Camera Book and composition targets
Performance measured after features are added	Performance budgets become part of art direction and acceptance criteria
More detail is used to improve quality	Composition, lighting, material consistency, hierarchy, and restraint are used to improve quality

2. Product and Emotional Design Principles

The new room pipeline should serve Keepsake's existing product laws rather than becoming a graphics project detached from the memory experience. Every visual decision should reinforce calm, tactility, personal history, and the primacy of photographs.

The room is the interface

Where practical, important functions should remain represented by physical objects: the scrapbook for memories, archive cabinet for stored photos, bookshelf for additional books, CRT for music, display case for meaningful physical artifacts, map/timeline for spatial and chronological memory, and letters or notes for selected notifications. Conventional UI remains available when accessibility, security, moderation, or efficiency requires it.

The House Has Memory

The room should preserve context and gradually develop history. A new room begins comparatively clean. Over months and years, the shelf fills, favorite books show use, artifacts appear, seasonal traditions return, photographs accumulate, and meaningful surfaces gain evidence of life. The goal is not to decorate a room into a personality on day one; memories slowly turn the room into one.

The House Has Manners

Keepsake should avoid interrupting editing, reading, or emotionally significant moments. Ambient events are sparse and context-aware. Silence is valid. One well-timed letter, CRT message, or returning seasonal object is preferable to a stack of badges and notifications.

The House Pays Attention

Atmospheric systems may respond to time, season, weather setting, current memory, or room state, but should do so subtly. The room may become warmer around a summer memory or quieter during Lights Out mode, but these responses should feel environmental rather than algorithmically performative.

Three layers of room content

Layer	Purpose	Examples
Permanent	Defines composition and visual identity	Architecture, desk, chair, archive, shelf, display case, CRT body, major lighting fixtures
Personal	Tells the owner's story	Photos, books, stickers, small keepsakes, scanned artifacts, CRT content, window-sill objects
Temporal	Tells us when we are	Seasonal decor, birthday letter, autumn leaves, holiday card, time-capsule envelope, temporary weather effects

Composition rule

Hero clutter instead of random clutter

Personalization should not mean arbitrary XYZ placement across every surface. Blender should define a small number of attractive clutter anchors on the desk, sill, shelf, case, and other surfaces. The application can fill those anchors according to ownership, memory relevance, season, and privacy. This preserves a professional composition even after years of use and greatly reduces overlap/collision problems.

- Desk Anchor A: camera and loose photographs.
- Desk Anchor B: pencils, markers, tape, or craft tools.
- Desk Anchor C: one personal keepsake.
- Desk Anchor D: letter, receipt, ticket, or small paper object.
- Desk Anchor E: seasonal or temporary object.
- Window-sill anchors: a few small objects only; never a continuous row of clutter.
- Display-case anchors: intentionally spaced shelf positions with explicit limits.

3. Visual Target: Illustrated Nostalgia

Illustrated Nostalgia is the proposed common art language for all four rooms. It should feel like a carefully art-directed indie game: believable enough to inhabit, stylized enough to feel authored, and nostalgic without becoming a literal period reconstruction.

Reference frame: Mirage demo



Reference qualities to preserve: warm localized desk light, cool exterior contrast, readable desk composition, believable prop density, tactile CRT presence, layered window depth, and a sense of a lived-in room.

The screenshot should not be copied literally. It functions as a quality and mood reference. The useful lesson is that perceived fidelity comes from composition, lighting, material response, silhouettes, and controlled clutter as much as from polygon count. Keepsake can pursue comparable perceived quality while remaining more stylized and more personal.

Target characteristics

- Softly stylized proportions with slight exaggeration on hero objects.
- Rounded/beveled edges where they catch light and improve readability.
- Materials that share a coherent roughness and color language instead of looking sourced from unrelated asset packs.
- Painterly or hand-authored exterior artwork rather than attempting to simulate an entire world in geometry.
- Warm localized interior lights contrasted with cooler exterior light at dusk/night.
- Evidence of life: straps, receipts, leaning books, worn edges, sun-faded patches, crooked frames, small marks, and asymmetry.
- Restrained animation. The room should feel alive, not busy.

- Photographs and scrapbook remain the visual protagonists.

Stylized exaggeration guidelines

Object	Suggested treatment	Reason
CRT	~105-110% visual emphasis	Iconic silhouette; nostalgic focal object
Scrapbook	~110% emphasis	Primary product object and emotional protagonist
Knobs/buttons/handles	~115-120% detail scale	Improves readability at authored camera distances
Books	Slightly thicker / clearer silhouettes	Reads better without dense geometry
Lamp shade / lantern	Slightly oversized	Strengthens lighting fiction and composition
Camera details	~110%	Makes a small nostalgic prop legible

Lighting hierarchy

Lighting should be designed around attention, not physical simulation alone. For Woodland, the default hierarchy should be: scrapbook first, desk objects second, window third, bookshelf/storage fourth, and the rest of the room fifth. Other rooms can change the practical sources while preserving a similarly deliberate hierarchy.

4. Four-Room Art Direction Matrix

The consolidation of Woodland and Snowy Mountain leaves four clearer, more differentiated room fantasies. They share memory functions, interaction conventions, and technical infrastructure, but should not look like palette swaps.

Room	Emotional fantasy	Interior signature	Exterior signature
Woodland Retreat	Secluded, familiar, reflective, seasonal	Warm timber, linen, paper, hearth/lantern warmth, aged tactile objects	Forest, hills/mountains, mist, seasonal transformations
Beachfront	Summer nostalgia, salt air, openness	Faded painted wood, canvas, ceramic, sea glass, sun-worn materials	Ocean/coast, bright horizon, moving cloud layers, sunset
Neon City	Late-night Y2K / future nostalgia	Graphite, plastic, alloy, CRT glow, controlled neon practicals	Layered skyline, rain, signs, distant traffic/light
Sky Castle	Dreamy escapism, impossible quiet home	Human clutter inside fantastical architecture; pearl/gold/stone/glass possibilities	Cloud ocean, floating islands, towers, bridges, waterfalls, giant sky

Shared skeleton: same functions, different fiction

The shared room skeleton should define functional slots rather than force identical models. A 'local light' slot can be a desk lamp in Woodland, a ceramic/brass lamp in Beachfront, a futuristic task light in Neon City, and a lantern or luminous fixture in another theme. The code sees a consistent function; the player sees a unique authored object.

System slot	Examples of implementation
Memory Core	Scrapbook, desk, chair
Storage	Bookshelf, archive cabinet, display case
Atmosphere	Window, primary/local light, secondary light, CRT
Memory Surfaces	Map, timeline, wall photographs or other curated surfaces
Personalization	Window sill, desk keepsake, stickers, books, artifacts, CRT content

Why Woodland and Snowy Mountain should be one room

Both concepts occupy the same emotional territory: a cozy nature retreat. Treating Snowy Mountain as Woodland's winter state produces stronger continuity, reduces duplicate art production, and turns the season system into a meaningful transformation rather than a cosmetic toggle. The user returns to the same desk, scrapbook, archive, and shelf while the world outside and selected interior details change.

Winter design goal

It should feel like the same familiar room on a snowy night, not a teleport to a different level.

5. Woodland Retreat 2.0: Seasonal Master Room

Woodland Retreat becomes the first production-quality vertical slice and the proving ground for the entire new pipeline. It should support four seasonal identities without replacing the underlying room.

Season	Exterior	Interior mood	Optional temporal details
Spring	Wet forest, mist, early green, blossoms	Soft daylight, cool damp window light	Small flowers, rain sounds, fresh paper/plant details
Summer	Deep green forest, warm distant hills	Warm daylight, long evening light	Open-feeling ambience, summer keepsake, insects/fireflies used sparingly
Autumn	Orange/rust canopy, haze, fallen leaves	Amber desk light, warm wood emphasis	A few leaves, seasonal card/object, richer textiles
Winter	Snow forest, alpine peaks, cold blue night	Hearth/lantern warmth against blue exterior	Fireplace, heavier textiles, snow on sill exterior, aurora on selected nights

Snowy Mountain systems worth preserving

- The warm-versus-cold lighting contrast.
- Fireplace/hearth as a winter practical and sound source.
- Lantern treatment as a seasonal/local-light variation.
- Aurora concept for selected winter nights.
- Snow atmosphere and alpine visual vocabulary.
- Positional fireplace ambience.
- Cabin detailing that can strengthen Woodland without creating a second complete room.

The current Snowy Mountain branch should remain intact as a reference/source branch. Its strongest ideas should be selectively migrated into Woodland 2.0. The large procedural alpine world, river, and distant terrain do not need to survive as real geometry if they can be represented more convincingly and cheaply in painted depth layers.

Seasonal asset-loading rule

Only assets required by the active season should be loaded. Winter-specific decor, textures, effects, and audio should not consume memory during summer. This principle should extend to all temporal systems and is central to the new performance strategy.

6. Procreate Window Art Specification

The project owner plans to create exterior artwork in Procreate on iPad. This is not a secondary decorative task; it can become one of Keepsake's signature art systems. The artwork should be created for parallax from the beginning rather than painted as one flattened image and cut apart afterward.

Recommended layer stack

Depth	Layer	Content	Motion expectation
0	Sky	Gradient, clouds, stars, moon/sun base	Essentially fixed or extremely slow
1	Far landscape	Mountains, distant islands, distant skyline	Very small parallax
2	Distant environment	Forest mass, coast, city blocks, castle silhouettes	Small parallax
3	Midground	Recognizable trees, structures, nearer terrain	Moderate parallax
4	Near environment	Large trunks, buildings, near cloud banks	Stronger parallax
5	Foreground	Branches, leaves, architectural edge elements	Strongest but still subtle
6	Window effects (often runtime)	Rain, snow, condensation, reflections, fog at glass	Screen/window-relative, not painted into base

Overscan

Every layer should extend beyond the visible window. Start with approximately 15-20% extra artwork around the intended composition. This prevents empty edges from appearing when layers shift during parallax. The final percentage can be reduced after the approved camera motion is known.

Paint behind occluders

A foreground tree must not hide an unpainted hole in the distant forest. When the foreground layer shifts, the background behind it becomes visible. Each deeper layer should therefore be complete enough to survive the maximum planned parallax movement.

Atmospheric perspective

- Far layers: softer edges, lower contrast, lower saturation, less detail, more influence from sky color.
- Near layers: stronger silhouettes, higher contrast, sharper edges, more texture and local color.
- Avoid equal detail everywhere; depth should be visible before any layer moves.
- Use broad value groups so the exterior remains readable behind glass and interior lighting.

Suggested Procreate organization

WOODLAND_WINDOW_MASTER
FOREGROUND

leaves
branches
MIDGROUND
trees
terrain
DISTANT_FOREST
forest mass
FAR_LANDSCAPE
hills
mountains
SKY
clouds
sky
REFERENCE
window-safe-area
camera-safe-area

Export guidance

- Keep the master Procreate file layered and non-destructive.
- Export each parallax group as a separate transparent PNG where transparency is required.
- Export fully opaque background/sky layers without unnecessary alpha where possible.
- Use consistent canvas dimensions and registration so layers align perfectly when imported.
- Name files predictably, e.g. woodland_autumn_00_sky.png, woodland_autumn_01_far.png, woodland_autumn_02_forest.png.
- Do not bake window frame, glass, rain, snow, reflections, or interior curtain into the base environment unless intentionally part of the painted world.
- Preserve a high-resolution master; generate optimized web derivatives separately.

Important artistic restraint

Parallax should create subconscious depth, not advertise itself. If the player notices layers sliding independently, the movement is probably too strong.

7. Inside / Outside / Threshold Architecture

A clean separation between physical interior, illustrated exterior, and the window threshold gives Keepsake a strong technical rule and helps prevent unnecessary geometry.

Zone	Primary technology	Examples
Inside	Blender + Three.js real geometry	Desk, scrapbook, chair, shelf, archive, display case, CRT, lamp, artifacts, photos
Outside	Procreate 2.5D + Three.js planes; optional Tesseract production experiments	Forest, ocean, skyline, mountains, cloud kingdom, distant terrain
Threshold	Hybrid Three.js / textures / selective motion assets	Glass, reflections, rain, snow, fog, curtains, sill, nearby branches, light entering room

The guiding test is simple: if the user cannot physically approach or meaningfully interact with a distant object, it probably does not need expensive mesh geometry. Spend geometry and shader complexity on objects that the user touches, reads, opens, collects, or remembers.

Room-specific exterior opportunities

Room	Painterly depth concept
Woodland Retreat	Seasonal forest, hills/mountains, near branches, mist; winter can introduce snow peaks and aurora.
Beachfront	Sky, horizon, distant water, nearer water bands, coast/rocks, foreground vegetation or railing; motion concentrated in water/clouds.
Neon City	Sky, distant skyline, mid skyline, near buildings/signage, rain/glass layer; animated signs can be isolated rather than making the whole city real-time.
Sky Castle	Sky, cloud ocean, distant floating islands, castle silhouettes, nearby structures, cloud banks, waterfalls; allows enormous scale at low runtime cost.

8. Figma as the Visual Constitution

Figma should become the shared visual source of truth before 3D production. It is not only for menus. It should define how the rooms look, how materials relate, how camera compositions are framed, and how the conventional UI beneath the diegetic room remains consistent.

Environment Bible deliverables

- One-page art-direction sheet for each of the four rooms before Woodland 2.0 modeling begins.
- Mood board and visual references.
- Color script for day, dusk, night, and key seasonal states.
- Material families: woods, fabrics, paper, plastics, metals, glass, ceramic, painted surfaces.
- Shape language and bevel/proportion rules.
- Camera composition boards with approved crops and focal hierarchy.
- Clutter density examples: too empty, target, too busy.
- Hero prop sheets for scrapbook, CRT, camera, lamp/lantern, archive, display case.
- Window safe area and Procreate layer planning frames.
- UI tokens for paper, ink, shadows, spacing, typography, lighting states, and focus states.
- Accessibility equivalents for room interactions.

Useful Figma techniques already identified

- Directional shadows for photographs, tickets, stickers, and paper artifacts.
- Progressive blur / glass treatments for plastic sleeves, CRT glass, and window concepts.
- 3D/isometric transformations for rapid prop and scrapbook composition studies before Blender.
- Variables/tokens for paper colors, wood families, room lighting, spacing, typography, and seasons.
- Auto Layout for settings, archive tools, toolbars, and responsive conventional UI.
- Text-on-path for handwritten labels and scrapbook decoration.
- Hierarchy, spacing, and contrast studies to keep the interface low-clutter.

The existing Keepsake Figma foundation should be expanded rather than replaced. The goal is to stop making aesthetic decisions independently in Blender, React, and runtime shaders.

9. Blender: Author the Physical Room

Blender should become the primary composition environment for the physical room. The strongest new production rule is: Blender owns placement; Three.js owns behavior.

Blender owns	Three.js owns
Appearance and silhouette	Visibility and state
Placement and resting pose	Interaction and hover/click
Scale and rotation	Opening/closing / state animation
Material assignment and UVs	Dynamic photo/scrapbook content
Bevels and proportions	Day/night/season selection
Camera composition targets	Camera transitions and runtime navigation
Static/baked lighting support	Dynamic practical-light state where needed
Interaction anchor placement	Permissions, memory state, personalization logic

Recommended Blender hierarchy

ROOM

Architecture

Window

Desk

CRT

Scrapbook

Camera

Markers

Printer

Lamp_or_LocalLight

Bookshelf

Archive_Cabinet

Display_Case

Chair

Rug

Wall_Art

Memory_Objects

Lighting_Anchors

Camera_Anchors

Interaction_Anchors

Asset families

Props should be designed as families rather than unrelated replacements. A Woodland CRT, clock, camera, cabinet, and lamp should share bevel language, roughness, color relationships, and period cues.

Beachfront receives its own family; Neon City and Sky Castle receive theirs. This is more important to cohesion than adding additional props.

Material library

- KS_Wood_WarmOak / KS_Wood_DarkOak
- KS_Fabric_Linen / selected room-specific fabric variants
- KS_Paper_Aged and clean paper variants
- KS_Plastic_Cream / dark translucent plastic where appropriate
- KS_Metal_Brass / painted metal / dark alloy families
- KS_Glass_CRT / window glass / display-case glass
- KS_Ceramic and small prop materials

Names are illustrative, but the principle matters: a controlled library reduces both visual inconsistency and material/draw-call proliferation.

10. Three.js / React Three Fiber Runtime Architecture

Three.js remains the correct runtime for the web-first Keepsake application, provided the room is engineered like a small game rather than treated as an ordinary decorative webpage.

Proposed runtime structure

RoomScene
 RoomShell
 VisibleZoneManager
 LightingRig
 InteractionLayer
 FurnitureLayer
 PersonalMemoryLayer
 WindowPortal
 Background
 FarLayer
 MidLayer
 NearLayer
 WeatherLayer

Semantic visibility zones

Frustum culling alone is not enough. Because Keepsake uses authored camera views, the application already knows what the player is looking at. It can explicitly turn unrelated systems off or down.

View	Desk	Window	Shelf	Archive	Map/Wall	Door/Guest
Scrapbook/ Desk	FULL	MEDIUM	LOW	LOW	OFF	OFF
Window	LOW	FULL	LOW	OFF	OFF	OFF
Shelf	LOW	LOW	FULL	MEDIUM	OFF	OFF
Archive	LOW	LOW	MEDIUM	FULL	OFF	OFF
Memory Wall	LOW	LOW	LOW	OFF	FULL	LOW

Render-on-demand philosophy

The current room uses continuous rendering during normal room view. Long term, Keepsake should move toward demand rendering where practical, invalidating frames for camera transitions, page turns, lighting changes, interactions, weather, CRT animation, and other visible motion. Continuous effects must be handled deliberately; switching to demand rendering blindly would break them. The goal is localized animation, not a permanently spinning render loop because one small object moves.

Camera Book

Every room should receive approved authored views: Establishing, Desk, Reading, Window, Shelf, Archive, Display Case, and Memory Wall. Each approved shot should record its focal hierarchy, visible-zone policy, and performance budget.

11. Mirage Tesseract: Defined Role

Tesseract should be treated as a production and experimentation tool, not as the foundation of Keepsake's runtime. Its useful role is to help create or prototype flat-layer depth, motion graphics, atmospheric loops, CRT/video content, environmental overlays, and composited visual ideas that can then be exported or reproduced efficiently in the Three.js experience.

Good Tesseract candidates

- Prototype parallax for a Procreate window painting before implementing the final runtime version.
- Subtle cloud or fog movement.
- Aurora studies for Woodland winter.
- CRT idle animations, title cards, music visualizers, or memory-video treatments.
- Rain/snow/environmental loops when a pre-rendered asset is preferable to a runtime simulation.
- Promotional cinematics and transitions.
- Screen-replacement and motion-graphics experiments.

What Tesseract should not own

- The interactive 3D room runtime.
- User permissions, scrapbook data, archives, accounts, routing, or application state.
- Core room geometry.
- A dependency required for the room to load.
- A proprietary runtime assumption that would make Keepsake fragile if Mirage pricing, APIs, or product direction changed.

Pipeline shorthand

Figma designs it -> Blender builds it -> Tesseract brings selected 2D elements to life -> Three.js performs it -> Keepsake remembers it.

12. Performance Findings from the Current Repository

The current repository already contains meaningful optimization work, but it also shows why the next improvement must be a production-pipeline change rather than simply adding more detail.

Room / branch	Observed asset data	Implication
Woodland balanced	~4.85 MB; 66,726 triangles; 73 materials; 266 primitives; max texture 512	Geometry is not extreme, but material count and layered runtime systems matter.
Beachfront balanced	~5.32 MB; 134,608 triangles; 80 materials; 309 primitives	Existing warm-cache profiling made this the first current optimization candidate.
Neon City	~3.08 MB recorded asset; prior export ~173,280 triangles / 312 primitives	Near geometry budget; improve composition/material hierarchy rather than adding geometry.
Sky Castle branch	~1.46 MB; 135,416 triangles; 230 meshes	Core asset is compact; procedural atmosphere can be replaced by painterly scale illusion.
Snowy Mountain branch	~6.35 MB; 166,785 triangles; 315 meshes; 324 primitives	Close to 180k triangle budget before considering runtime atmosphere; strong case for consolidation and 2.5D exterior.

Existing warm-cache performance measurements

Room	Ready	Warm initial view	p95 frame	Draws	Visible triangles
Woodland	2.57 s	85 fps	14.3 ms	485	138,636
Beachfront	2.71 s	71 fps	16.7 ms	552	277,926
Neon City	2.31 s	93 fps	12.5 ms	487	266,780

These measurements were local warmed-cache results and should not be treated as cold-network or mobile guarantees. They are still useful because they show that draw calls and visible scene complexity are already substantial.

Material consolidation

Woodland currently records 73 materials and Beachfront 80. The new pipeline should reduce material fragmentation through shared material families and atlasing where appropriate. The goal is not an arbitrary universal material count, but materially fewer state changes and a more coherent visual language.

Room asset cache concern

The current room asset cache retains loaded source GLTFs for the session. Cloned materials are disposed on unmount, but cached source geometry/textures remain available. That behavior may be intentional for

fast revisits, but it is a high-value profiling target because repeated switching among high-quality rooms can retain multiple asset sets in memory. This should be measured before blaming any single system for crashes.

13. Performance Budgets and Quality Tiers

Performance budgets should be treated as design constraints, not cleanup tasks. The following are project targets for iteration and should be validated on representative devices.

Metric	Target direction
Visible triangles	Approximately 80k-150k in normal authored views where practical
Draw calls	Target <250 in common room views
Hero textures	1K-2K where the object visibly benefits
Minor prop textures	256-512 where sufficient
Exterior	2.5D layered artwork by default
Dynamic shadow lights	Prefer one primary shadow-casting source; add local lights sparingly
Desktop DPR	Roughly 1.25-1.75 depending on quality/device
Mobile DPR	Roughly 1.0-1.25
Desktop frame target	60+ FPS on target hardware
Mobile frame target	30-60 FPS with intentional reduced tier

Quality tiers should change more than DPR

Tier	Expected differences
Desktop High	Higher DPR, selected shadows, higher texture tier, fuller parallax/atmosphere, selected reflections/effects
Desktop Balanced	Moderate DPR, baked/static lighting emphasized, reduced dynamic effects, same approved composition
Mobile	Lower texture tier, fewer particles, reduced shadows/reflections/post, simplified decorative props, same interaction and visual hierarchy

Progressive loading

1. Load application shell and essential UI.
2. Load basic room shell and approved camera.
3. Load desk + scrapbook so the primary interaction becomes available quickly.
4. Load window background/parallax.
5. Load shelf/archive/display case only when relevant to the active view or shortly before transition.
6. Load high-cost user artifacts only when the display case or linked memory requires them.

14. Current Procedural Scenery: What to Keep and What to Replace

System	Current strength	New direction
Woodland scenery	Demonstrates trees, mist, weather, seasonal ambience	Replace most distant geometry with Procreate depth layers; retain only threshold effects that benefit from runtime behavior.
Beachfront scenery	Sky/water/horizon/cloud/boat/weather already separated conceptually	Move distant coast/ocean composition toward painterly layers; keep only motion that adds value.
Sky Castle scenery	Strong fantasy scale: cloud sea, waterfalls, stars, day/dusk/night	Make it the most painterly room; use layers to imply enormous world scale; keep selective waterfall/cloud motion.
Snowy Mountain scenery	Excellent warm/cold contrast, aurora, river, stars	Absorb into Woodland winter; replace distant river/mountains with illustrated layers; retain hearth/aurora concepts selectively.
Neon atmosphere	Bespoke shaders and distinctive holographic/night identity	Keep as a special-case room but reduce simultaneous effects and prioritize composition over additional spectacle.

Animation restraint

The room should not behave like a technology demo. At night in Woodland, a small branch shift, slow cloud movement, subtle CRT flicker, quiet lamp glow, and occasional weather can be enough. If trees, fog, particles, curtains, lights, reflections, CRT, fireplace, and props all move continuously, nostalgia turns into visual noise and battery/GPU cost rises at the same time.

15. Artifact Display Case and Memory Objects

The display case remains an important part of the new room architecture because it gives physical memories a home distinct from paper memories. It should be authored as part of every room's composition, with room-specific materials but the same underlying interaction model.

Memory Box	Display Case
Paper / mostly flat memories	Physical / dimensional memories
Tickets, letters, newspaper clippings, drawings, postcards, programs	Toys, trophies, souvenirs, figurines, sports memorabilia, heirlooms, vacation objects
Can be scanned/photographed and placed on scrapbook pages	Can be represented by optimized 3D models and linked back to memories

Phone-to-case 3D scan workflow

- 7. User chooses Add Artifact.
- 8. Phone captures photos and/or depth around the object.
- 9. An external reconstruction/native scanning workflow creates the initial 3D model.
- 10. Automatic cleanup removes background, corrects orientation, simplifies geometry, optimizes textures, and establishes scale.
- 11. A lightweight GLB/gLTF derivative is created for web display while the source is preserved separately.
- 12. User previews the artifact and chooses an authored shelf position.
- 13. Artifact becomes a first-class memory record with title, year, description, privacy, provenance, and links to scrapbook memories.

Artifact as navigation

A display-case object should not merely decorate the room. Selecting a childhood baseball, concert wristband, or souvenir can reveal a short environmental note and offer a Memory Trail into the related scrapbook page. This turns the room into a reverse index of the user's life.

Privacy

Private artifacts should not be rendered for unauthorized visitors at all. Permission inheritance should apply to the artifact asset itself, its metadata, and its linked memories. A visitor may be allowed to see the case but not inspect certain objects.

16. Scrapbook and Room Integration

The room redesign should not distract from the scrapbook editor. The scrapbook remains the product's core. The room is successful when it makes opening the book feel meaningful and when memories visibly influence the space over time.

Generic PageElement architecture

PageElement

id

pageId

type: photo | artifact | sticker | caption | keepsake

sourceId

x / y

width / height / scale

rotation

zIndex

frontBackState

visibility

locked

createdAt

updatedAt

This model lets photos, tickets, clippings, drawings, letters, stickers, captions, and future keepsakes share the same drag/rotate/resize/layer/undo/persistence infrastructure. It also makes room artifacts and scrapbook artifacts part of one memory system rather than separate gimmicks.

Photograph backs and physicality

Photographs may flip to reveal handwritten notes, date, place, people, longer memory text, and selected metadata. Exact GPS or sensitive EXIF should never be exposed automatically. This feature fits the tactile art direction and gives the room/scrapbook system a physical logic beyond a standard gallery.

17. Seasonal Room Memory and Long-Term Aging

Seasonal Room Memory becomes substantially more important after the Woodland/Snowy consolidation. The season can now coordinate exterior art, light, sound, weather possibilities, textiles, small decor, and returning traditions.

What may return

- A holiday card or letter placed in a previous year.
- A small decoration the owner chose to keep as a seasonal tradition.
- A winter blanket or object in an authored clutter anchor.
- A selected seasonal CRT idle screen or music association.
- A window-sill item associated with that season.
- A subtle change in shelf or desk history, provided the clutter budget is respected.

What should not happen

- Do not repopulate the room with every historical seasonal item.
- Do not create urgency, streaks, or FOMO around seasons.
- Do not silently change user-authored scrapbook content.
- Do not allow seasonal decoration to obscure photographs or core interactions.
- Do not require live weather; season and ambience can remain intentionally authored.

18. Camera and Interaction Direction

Authored viewpoints are now the default. This is both an artistic and technical decision. It allows the team to compose every frame, guarantee important objects are readable, control what must be loaded, and spend performance budget where it is visible.

Recommended standard views

View	Primary purpose
Establishing	Enter the room and understand its identity.
Desk	Interact with scrapbook, CRT, camera, small desk memories.
Reading	Close, emotionally focused scrapbook view.
Window	Enjoy season/weather/exterior and related settings.
Shelf	Choose books and browse long-term collection.
Archive	Access unplaced/older photos and categories.
Display Case	Inspect physical artifacts and linked memories.
Memory Wall	Timeline/map/wall memories with room-specific composition.

WASD/free-roam remains a future option. The art pipeline should not make it impossible, but it should not force every wall, floor edge, and backside of every prop to be production-ready today.

19. Audio and Room Sound Geography

Audio should reinforce the authored zones without becoming required for understanding. The current positional fireplace work in Snowy Mountain is a useful precedent.

Zone	Examples
Window	Rain, wind, birds, distant ocean, city hum, high-altitude wind
Desk	Paper movement, page turns, pencil/tape sounds, subtle object handling
CRT	Music and screen hum localized to the device
Archive	Drawer, folders, paper stack movement
Fireplace / winter Woodland	Crackle and low room warmth ambience
Display Case	Very restrained glass/door/object handling sounds

Use smooth crossfades and independent ambience/music/interaction volume controls. Essential information must never be conveyed by sound alone.

20. Implementation Roadmap

Phase 0 - Freeze room feature growth

Temporarily stop adding new decorative room systems. Preserve the current branches and use them as references. This prevents the art-direction rewrite from chasing a moving target.

Phase 1 - Four-room Figma matrix

- Create one-page art-direction sheet for Woodland, Beachfront, Neon City, and Sky Castle.
- Define shared skeleton, room-specific material families, signature lighting, exterior concept, and hero props.
- Approve Illustrated Nostalgia style target and clutter density.
- Create initial Camera Book frames for all four rooms so Woodland is not designed in isolation.

Phase 2 - Woodland 2.0 pre-production

- Detailed Woodland mood board and color script.
- Spring/summer/autumn/winter transformation sheet.
- Desk/scrapbook focal hierarchy.
- Material library.
- Clutter anchors.
- Window safe area and Procreate template.
- Lighting states: day, dusk, night, rain, winter/hearth.
- Performance budget and acceptance checklist.

Phase 3 - Blender blackout

Build architecture and hero furniture in graybox first: walls, desk, chair, window, shelf, archive, display case, CRT, scrapbook, rug, lamp/local light. Approve proportions and camera compositions before detail work.

Phase 4 - Parallax window prototype

14. Create one Woodland illustration in Procreate.
15. Split/export sky, far, forest, mid, near, foreground layers.
16. Prototype depth and motion.
17. Implement a reusable ParallaxWindow component in Three.js.
18. Add one weather effect and one dusk/night grade.
19. Measure visual quality and runtime cost on desktop and phone.

Phase 5 - Hero assets and material pass

- Scrapbook
- CRT
- Camera
- Printer if retained
- Clock
- Lamp/lantern

- Chair
- Archive cabinet
- Bookshelf
- Display case

Phase 6 - Lighting and cinematography

Create controlled day/dusk/night/winter lighting variants and compare screenshots from every Camera Book view. The acceptance question is not merely 'is this realistic?' but 'does this make me want to sit here and open the scrapbook?'

Phase 7 - Optimization

- Consolidate materials and atlases where appropriate.
- Compress textures and GLBs.
- Add LODs only where they create measurable value.
- Implement semantic visibility zones.
- Lazy-load non-visible props and user artifacts.
- Reduce shadow casters and dynamic lights.
- Use demand/localized rendering where appropriate.
- Profile room switching and source-asset cache behavior.
- Test on representative phone and laptop hardware.

Phase 8 - Propagate the proven pipeline

After Woodland 2.0 reaches the visual and performance target, apply the same production system to Beachfront, Neon City, and Sky Castle. Do not rebuild all rooms simultaneously.

21. Room-by-Room Proof Goals

Room	What it proves	Key challenge
Woodland 2.0	Entire production pipeline and four-season continuity	Cohesion, performance, parallax, memory-first composition
Beachfront 2.0	Environmental motion and bright atmospheric rendering	Water/cloud motion without turning the exterior into expensive full 3D
Neon City 2.0	Controlled emissive/night rendering	Prevent neon, holograms, shaders, reflections, and CRT effects from competing
Sky Castle 2.0	Painterly world-building at impossible scale	Keep the interior human and memory-focused while the exterior is fantastical

Sky Castle special guidance

Sky Castle should probably be the most painterly of the four rooms. Its current branch already demonstrates cloud sea, waterfalls, stars, and a larger far plane. The new approach can make the world feel vastly larger through illustration rather than increasing camera range and geometry. The interior should remain recognizably Keepsake: worn scrapbook, mug, photos, books, CRT, letters, and artifacts inside impossible architecture.

22. Acceptance Criteria for a 'Professional' Keepsake Room

A room should not be considered finished because all intended props are present. It should pass a combined art, product, and performance review.

- Every approved Camera Book view has a clear focal hierarchy.
- The scrapbook is visually important whenever it is supposed to be.
- No major prop looks like it belongs to a different asset pack.
- Materials share a coherent roughness, edge, and color language.
- Lighting communicates the room's emotional identity and time/season.
- Exterior depth is convincing without drawing attention to the parallax trick.
- Personal clutter fits authored anchors and never destroys composition.
- The room remains legible when personal content is sparse and when it has accumulated.
- Important interactions are discoverable without floating UI clutter.
- Reduced-motion and accessibility alternatives remain functional.
- Common views meet agreed draw-call/frame-time/memory targets on representative devices.
- Room switching does not cause runaway memory growth.
- Mobile quality is intentionally art-directed rather than simply 'desktop with effects missing'.
- A screenshot of the room is recognizable as Keepsake without seeing the logo.

23. Open Questions to Resolve During Pre-Production

- Final name for the consolidated Woodland room (Woodland Retreat is a working title).
- Whether the fireplace is always physically present in Woodland or appears/activates only in winter.
- How many seasonal interior swaps are acceptable before continuity begins to feel like a different room.
- Exact Procreate master canvas dimensions and texture derivatives after the first camera/window prototype.
- Whether day/night uses a shared painted environment with grading or dedicated sky/environment variants for the highest-quality states.
- Which room-specific hero objects are allowed to deviate from the shared skeleton.
- Final desktop/mobile performance budgets after testing on target devices.
- Cache eviction strategy for room source assets after profiling real room-switch memory behavior.
- How much of the current runtime furniture replacement system can be retired once Blender becomes authoritative.
- Which Tesseract-produced assets, if any, offer enough value to ship versus recreating the effect directly in Three.js.
- Whether launch should include all four rooms or prioritize fewer rooms at a higher finish level.

Quality principle

Three unforgettable rooms are better than five rooms where two still feel like prototypes. The four-room plan is a production target, not a requirement that every environment ship simultaneously.

24. Astra Workflow Handoff: Source-of-Truth Summary

The following section is intentionally explicit so another agent/workflow can act on the plan without reconstructing the conversation.

20. Do not add new room features until the art-direction pipeline is established.
21. Treat the current repository rooms and experimental branches as reference implementations, not final visual architecture.
22. Reduce selectable room concepts to Woodland Retreat, Beachfront, Neon City, and Sky Castle.
23. Absorb Snowy Mountain into Woodland as the winter transformation. Preserve its branch for source/reference work.
24. Create a Figma four-room art-direction matrix before rebuilding Woodland.
25. Use Illustrated Nostalgia as the shared visual target: stylized cinematic realism, painterly atmosphere, tactile hero props, restrained clutter and motion.
26. Use authored camera viewpoints as the primary navigation model. Do not design around WASD/free-roam yet.
27. Remove major furniture customization. Each room owns its desk/chair/cabinet/etc.; user expression happens through smaller memory-bearing objects.
28. Make Blender authoritative for room composition, hero furniture, static placement, material assignment, and camera anchors.
29. Make Three.js authoritative for interaction, dynamic state, camera transitions, seasonal state, personalized content, permissions, and runtime presentation.
30. Create Procreate exterior artwork as layered 2.5D scenes with overscan and complete background coverage behind occluders.
31. Treat exterior distant worlds as illustrated illusions unless the user can meaningfully interact with them.
32. Keep threshold effects such as glass, rain, snow, reflections, nearby branches, curtains, and incoming light as hybrid runtime elements.
33. Use Tesseract only as an optional production/prototyping tool for parallax, motion graphics, atmospheric loops, CRT content, and related assets.
34. Implement semantic visibility zones tied to the authored Camera Book so unrelated systems can be disabled or downgraded.
35. Create real performance tiers for Desktop High, Desktop Balanced, and Mobile; do not rely only on DPR differences.
36. Profile room-switch memory and the existing GLTF cache before making a major engine migration.
37. Use Woodland 2.0 as the vertical slice. Prove the entire pipeline there before rebuilding the other rooms.
38. Preserve the scrapbook as the visual and product protagonist. A beautiful room that distracts from memory creation is a failed room.

Final north-star statement

Keepsake room vision

A Keepsake room should feel like a place the user has lived with, not a scene they configured. Its architecture is authored, its atmosphere is cinematic, its outside world is painterly and suggestive, and its personality comes from the memories that slowly accumulate inside it.